

Yutaro Kiyota

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Github: <https://github.com/ykydh/Portfolio>

PROJECTS

Stock Prediction Platform - *Python, TypeScript, FastAPI, SQL, React, XGBoost*

- Built an end-to-end full-stack stock prediction system leveraging FinBERT (NLP) for financial news sentiment analysis and XGBoost regression to predict next-day stock price movement using time-series and sentiment-based features
- Designed and implemented a relational SQL database schema to store historical stock prices, news sentiment, and model outputs, ensuring data integrity, efficient indexing, and low-latency queries
- Developed a REST API using FastAPI to serve real-time prediction and confidence metrics and integrated it with a TypeScript/React frontend for interactive data visualization

Socket Programming Project - *C++, TCP/IP, Networking*

- Engineered a TCP socket communication system in C++ with a custom packet framing protocol, enabling reliable data transmission using 4096-byte buffered messages with size-prefixed headers
- Implemented a modular client-server architecture, abstracting socket initialization, connection handling, and data exchange into reusable, maintainable classes
- Applied low-level networking concepts including byte streams, buffering, and error handling to ensure robustness and scalability

EXPERIENCE

Missouri University of Science & Technology - Satellite Research Team

Software Developer | *Aug 2023 - Present*

- Designed and extended an Operating System Abstraction Layer (OSAL) over FreeRTOS to

standardize file system and task-level interactions for CubeSat flight software

- Designed and debugged a multithreaded Ethernet communication module, supporting concurrent data transfer between satellite subsystems
- Designed and improved Operating System Abstracting Layer of freeRTOS for a satellite flight software’s file system to ensure smooth execution of file system communication between application and OS.

Missouri University of Science & Technology - Residential Life

Senior Resident Assistant | *Aug 2024 - Present*

- Supported residents facing academic and mental health challenges by providing resources, mentorship, and empathetic communication
- Organized and executed wide variety of community-building programs to improve engagement and foster a inclusive community
- Mentored and trained first-year resident assistants, fostering leadership, collaboration, and conflict-resolution skills by introducing individually tailored approaches

SKILLS

C++ Python Assembly FastAPI
React Git SQL (database design, queries)
Leadership, Mentorship Communication

EDUCATION

Bachelor of Science in Computer Science

Aug 2023 -Present

Missouri University of Science & Technology

GPA: 4.0 / 4.0 | Dean’s List